Tiago Ferrão-Lindqvist

Data Engineering | Test Data Automation

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PROFESSIONAL SUMMARY

Mechanical engineer with 12 years of experience in product design, testing and validation, and engineering data analysis. Builds Python tools to process test-rig data, check results against specifications, and generate reports. Developed an internal processing tool used by four test engineers and a Power BI dashboard used by around 15 colleagues. Experienced with Python, SQL Server, MATLAB/Simulink and test data; currently completing an online data engineering course.

TECHNICAL SKILLS

Programming and data: Python (pandas, NumPy, Matplotlib, npTDMS), SQL, MATLAB/Simulink, Excel/VBA
Data tools: SQL Server, Power BI, SQLite, Flask
Engineering and testing: ANSYS Mechanical, SolidWorks, CATIA V5, LabVIEW, Minitab, design of experiments (DOE), GD&T, DFMEA, 8D problem solving
Development tools: Git (personal projects), Docker Compose (tutorial project)

EXPERIENCE

Test & Validation Engineer | Corvane Automotive Components | Setúbal
Mar 2019–Present

- Plan and run durability, vibration and climatic tests on seat-adjuster mechanisms across six servo-hydraulic and electric test rigs; prepare reports for OEM customers.
- Develop Python scripts to parse TDMS and CSV rig outputs (around 2 GB per week), check channels against test specifications, and generate standard report figures. Reduced report preparation from about three days to half a day per test.
- Maintain an internal data-processing tool scheduled nightly on a laboratory PC. It writes summary results to a staging table in the plant's SQL Server database and is used by four other test engineers.
- Query the plant's test-results database for trend analysis across three product generations.
- Built a Power BI dashboard using the staging table to show pass/fail trends and cycles to failure by product generation; used by around 15 validation and design engineers.
- Correlate rig results with FEA in ANSYS Mechanical and lead 8D root-cause analysis on field failures. Proposed a redesigned adjuster rail bracket that cut warranty returns for the part by 40%.
- Maintain MATLAB/Simulink models of the adjuster drive, used to select rig load profiles that match customer duty cycles.
- Trained three apprentices in rig safety and test setup (2022–2024).

Mechanical Design Engineer | Halvik Pumps | Porto
Sep 2014–Feb 2019

- Designed centrifugal pump casings and impellers in SolidWorks and produced manufacturing drawings with GD&T; owned DFMEA for two product lines.
- Performed structural FEA of casings in ANSYS. Supported a move from grey cast iron to ductile iron for one casing family, saving 12% in material cost.
- Wrote a LabVIEW data-acquisition program for the pump performance test bench, reducing test setup time from about two hours to 30 minutes.
- Supported conversion of the performance test bench to a Siemens S7 PLC-controlled setup, changing setpoints and timers in TIA Portal under the automation engineer's supervision.

SELECTED DATA PROJECTS

Bike-sharing Trips API | Personal learning project | 2025

- Built a small Flask API over a public bike-sharing dataset loaded into SQLite, with three read-only endpoints.
- Followed a tutorial to run the project with Docker Compose; maintains personal project repositories using Git.

EDUCATION & PROFESSIONAL DEVELOPMENT

MSc Mechanical Engineering | Sado Valley Polytechnic | 2014
Thesis: Fatigue life prediction of welded brackets under variable-amplitude loading

Online Data Engineering Course | In progress
Completed 6 of 9 modules

LANGUAGES

Portuguese (native) | English (fluent) | Spanish (good)